Live Liquidity for LMSR Prediction Markets

Fair Fees, Safe Reserves, and Less Entry-Time Bias

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Abstract

Logarithmic Market Scoring Rule (LMSR) prediction markets provide continuous prices, guaranteed liquidity, and bounded sponsor loss in one mechanism [1]. On-chain LMSR systems have usually kept that structure sponsor-funded rather than open to live liquidity-provider (LP) entry [4, 5]. This paper studies a constrained alternative: a multi-outcome LMSR in which LPs may enter during active trading, may not withdraw before settlement, earn only fees collected after they join, and receive residual collateral only after fixed winner claims are fully reserved.

We study that mechanism through three linked models: an exact high-precision benchmark based on parallel LMSR components, a compressed global-state candidate, and a fixed-point model oriented toward the Algorand Virtual Machine (AVM), the smart-contract environment used by question.market. Across deterministic, Monte Carlo, adversarial, and boundary scenario families, we evaluate price continuity, slippage, LP fairness by entry time, aggregate reserve coverage, cohort sponsor-solvency diagnostics, required order invariance, and divergence between exact and simplified models.

The main findings are clear. In high precision, the compressed Layer B model matches the exact benchmark essentially perfectly. Prospective-fee fairness is straightforward to enforce. Reserve-based residual release preserves winner seniority while allowing LP capital to unlock before every winner has claimed. The hard economic issue is residual fairness: naive residual pooling favors later LPs, fully linear time-weighting overcorrects, and a mild affine rule performs materially better. A first-stage event-step calibration already reveals that structure. A stronger held-out study that rescales the same logical paths across short, medium, and long scenario clocks then selects a lifetime-normalized affine premium $\lambda_{\text{norm}} = 0.150$. On held-out ordinary-regime packs, that rule reduces the mean late-minus-early NAV-per-deposit gap to about -0.0068, the mean absolute gap to about 0.0394, and the duration-bucket drift to about 0.0223, while the best event-step alternative remains badly clock-sensitive. In a matched resolved-only comparison against a Gnosis-style fixed-product market maker (FPMM) baseline, the calibrated active-LP LMSR is more neutral overall.

The validated ordinary operating regime supports bounded implementation. Production deployment should enforce explicit late-entry skew gates and low-tail many-outcome limits, because those states define the main validated boundary.

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1 Introduction

LMSR prediction markets do something unusual. They turn a live book of positions into a continuous probability surface while keeping the market maker's loss bounded. That is why LMSR remains one of the cleanest mechanisms ever proposed for prediction markets [1]. It gives a market constant liquidity without requiring a matching counterparty for every trade, and it does so in a form that still works for more than two outcomes. In the standard presentation, however, LMSR is backed by a sponsor or creator who funds the market maker up front. Early on-chain implementations such as Gnosis followed that sponsor-centered structure as well [5]. The sponsor bears the market-maker risk, traders take outcome positions, and the fee or residual economics belong to the sponsor side rather than to an open pool of liquidity providers.

This property matters even more in crypto markets because it addresses the bootstrapping problem directly. Order-book prediction markets such as Polymarket's central limit order book (CLOB) can provide excellent price discovery once bids and asks are resting in the book, but they face a cold start: before traders post compatible orders, there is no immediate executable liquidity and no meaningful spread to trade against [7]. An AMM solves that by quoting from committed capital from the first trade onward. In prediction markets, that is especially valuable because many markets begin thin, event-specific, and informationally sparse. The market needs a way to open before a dense two-sided book exists.

That framing is mechanically clean, but it has an obvious limitation. Market depth is mostly fixed at launch. If a market later attracts meaningful participation, it is natural to ask whether outside capital should be able to enter and deepen the curve. If that can be done without moving the current price, the result is powerful: liquidity can grow in place. In ordinary decentralized-finance (DeFi) markets, this question is straightforward because LPs deposit against an inventory pool with comparatively simple accounting. In LMSR prediction markets, it is not. The market maker does not merely hold assets. It carries contingent liabilities against future winning claims, and those liabilities depend on both the current price vector and the entire trade path that produced it.

That difference matters for risk as well as for accounting. In a fixed-product AMM, LPs often think in terms of inventory drift and impermanent loss relative to holding the underlying assets outright. That is not the main exposure in the active-LP LMSR studied here. These LPs are not primarily warehousing a token basket that can be withdrawn on demand. They are posting locked sponsor capital against contingent winner claims, earning only prospective fees, and holding a junior claim on whatever residual value remains after senior obligations are fully reserved. The relevant tradeoffs are therefore lockup risk, timing risk, and junior-residual sponsor risk rather than the usual constant-product impermanent-loss story.

This makes live LP entry a mechanism-design problem rather than a simple add-liquidity function. If a late LP enters an already skewed market, what exactly are they buying? A share of future fees is easy to define. A share of terminal residual value is much harder. This is the core intuition the paper takes seriously: later LPs have an informational timing advantage. They are entering after part of the uncertainty has already been resolved by order flow. If the mechanism ignores that, it induces a timing subsidy; if it overcorrects, it makes late capital unattractive exactly when deeper markets would otherwise be useful. If LP withdrawals are also allowed before settlement, the accounting problem becomes much worse because trader sell rights and winner claims can be impaired by sponsor exit.

The contribution of this paper is practical. A live-entry, no-withdrawal active-LP LMSR can be made coherent enough for implementation if three conditions hold. First, fee fairness must be strict: new LPs should only earn fees collected after they join. Second, winner claims must remain fixed and senior. Third, residual release must use explicit reserve accounting rather than forcing LP capital to wait for every winner to claim.

We evaluate that thesis through simulation rather than intuition alone. The evidence suggests that the active-LP direction is viable, that fee fairness is the clean backbone of the model, and that residual bias is measurable and materially reducible rather than fatal. More specifically, the simulations

show that the naive intuition about late-LP advantage is correct, that a first-stage event-step fit is not portable enough to trust on its own, and that a normalized-exposure correction generalizes much better across held-out duration-scaled workloads without breaking the rest of the mechanism. The evidence also shows where the limits remain: low-tail, many-outcome regimes are still the main deployment boundary.

The paper makes four claims. First, live LP entry can deepen an LMSR without breaking price continuity. Second, prospective fee fairness is straightforward to enforce. Third, residual fairness is the real economic problem, but a mild lifetime-normalized affine rule materially reduces timing bias. Fourth, the mechanism is viable only inside an explicit operating envelope, with high-skew late entry and low-tail many-outcome states treated as boundaries rather than ignored tails.

2 Why Live Liquidity Is Hard

2.1 Why LMSR Usually Has a Sponsor

LMSR is a cost-function market maker. A market-maker state induces a price vector over outcomes, and trades move that state while paying or receiving amounts determined by the cost function. The central parameter controlling local depth is conventionally written as b or B. Larger depth reduces slippage but increases the amount of collateral required to support the market-maker's bounded-loss property [1, 2, 3, 4].

In the canonical LMSR framing, this collateral is sponsor capital. The sponsor is not "just another trader." A trader owns outcome claims. A sponsor owns the economics of the market-maker side: fee flow and whatever junior equity remains after trader claims are paid. This distinction is easy to blur in informal discussion because a complete-set trade superficially resembles broad market exposure. Economically, though, it is different. Buying outcome claims does not make the trader the owner of the market maker's fee stream or its contingent liabilities.

Early on-chain LMSR implementations followed this sponsor-centered structure directly. Gnosis' `StandardMarket` allows creator funding and creator-side fee withdrawal, while its `Campaign` contract allows multiple contributors to crowdfund a market before launch rather than providing continuous live LP entry during active trading [5, 6]. This is the closest implementation precedent for our work, but it stops short of the active-LP mechanism studied here.

2.2 Minimal LMSR Definitions

The standard LMSR cost function is

$$C(\mathbf{q}) = B \ln \left(\sum_{i=1}^{N} e^{q_i/B} \right),$$

where $\mathbf{q} = (q_1, \ldots, q_N)$ is the market-maker inventory state and B is the liquidity parameter. The corresponding instantaneous price of outcome j is

$$p_j(\mathbf{q}) = \frac{e^{q_j/B}}{\sum_{i=1}^{N} e^{q_i/B}}.$$

These prices always sum to one. A larger B makes the market deeper by reducing local slippage. In the neutral N-outcome start state, LMSR's bounded-loss property implies a worst-case sponsor loss of $B \ln N$ [1, 2]. This is why the sponsor role matters: someone must capitalize the market maker whose quotes the traders are interacting with.

The fee side of our mechanism uses a cumulative fee index. If trade t contributes fee flow F_t to LPs and the active sponsor-share supply at that instant is S_t, the fee index updates as

$$I_t = I_{t-1} + \frac{F_t}{S_t},$$

and a cohort with share units s_k that entered when the index was $I_{k,\text{entry}}$ has claimable fees

$$\text{fees}_k(t) = s_k \left(I_t - I_{k,\text{entry}}\right).$$

This is the formal statement of the prospective-fee rule: a later LP dilutes earlier LPs only on fee flow that occurs after joining.

2.3 Why live LP entry is hard

Three problems appear immediately when LPs are allowed to enter after launch.

The first is price continuity at entry. Adding LP capital should deepen the market, but it should not move the current price vector merely because capital arrived. If LP entry itself changes prices, the mechanism creates artificial mark-to-market transfers even before any informational trade occurs.

The second is fee fairness. This problem is comparatively tractable. If a new LP joins at time t, it should receive no share of fees collected before t and should dilute earlier LPs only on fees collected after t.

The third is residual fairness. This is the hard part. If residual value is distributed by a naive pooled-share rule, later LPs may buy into sponsor equity too cheaply. If residual accounting is too rigidly cohort-isolated, the model becomes difficult to implement and may not even reflect the correct pooled-risk interpretation of the sponsor side. The challenge is to find a rule that preserves winner seniority, prevents obvious extraction, and remains simple enough to implement on-chain.

This challenge is not unique to our model. Othman and Sandholm show that no market maker can simultaneously be no-arbitrage, path-independent, and liquidity-sensitive, and they explicitly warn that naive schemes which feed transaction revenue back into LMSR liquidity can distort prices or break path independence [10]. The active-LP problem can therefore not be reduced to a generic story of "just increasing b." It is a constrained contingent-liability design problem.

The design choice in this paper is explicit. We do not attempt liquidity sensitivity in the Othman-Sandholm sense where liquidity endogenously changes as a function of market evolution or fee recycling. We keep the LMSR pricing rule path-independent between LP events and allow depth to change only through explicit sponsor entry. In other words, the mechanism gives up endogenous liquidity sensitivity in exchange for cleaner accounting around price continuity and settlement seniority.

2.4 Design Boundaries

This paper adopts the following design envelope.

- LPs may enter during `ACTIVE`.
- Markets must be bootstrapped by an initial sponsor or sponsor pool before active LP entry begins.
- LPs may not withdraw during `ACTIVE`.
- Fee rights are strictly prospective.
- Winner claims are fixed and senior rather than parimutuel.
- Residual release is reserve-based: the protocol first reserves the amount still required to honor outstanding winner or refund claims, and only the free pool above that reserve can be claimed by LPs.
- Markets may have more than two outcomes.
- The mechanism is judged by explicit simulation evidence first, not by informal intuition.

These constraints are deliberately narrow. The aim is not to solve every possible LP design for prediction markets, but to identify one design that is economically defensible and operationally plausible enough to justify implementation.

2.5 Related Work

There are at least four relevant lines of prior art.

The first is the original market-scoring-rule and LMSR literature. Hanson motivates automated market makers as market scoring rules and argues for the logarithmic rule as a uniquely local way to preserve conditional structure under trading [1]. Chen and Pennock then place bounded-loss market makers into a broader utility and cost-function framework, while Abernethy, Chen, and Vaughan formalize efficient automated market making through convex optimization and connect it directly to online learning [2, 3]. Together these papers form the theoretical spine for any LMSR-based prediction-market design.

The second is early on-chain LMSR practice. Gnosis' first-generation prediction-market contracts provide a concrete blockchain implementation of creator-funded LMSR markets, with creator-only funding and fee withdrawal in `StandardMarket`, and a separate `Campaign` mechanism for crowd-funded market launch rather than continuous live LP entry [5, 6]. This is the closest direct sponsor-funded implementation precedent for our work, but it stops short of the active-LP mechanism studied here.

The closest deployed *active-LP* comparator is not LMSR but the Gnosis fixed-product market maker (FPMM) built for conditional tokens. In the Gnosis AMM repository, FPMM is documented as a prediction-market AMM for conditional tokens and implemented concretely in `FixedProductMarketMaker.sol` [8, 9]. The mechanism supports permissionless funding, share burning, fee withdrawal, and transferable LP shares, while tracking a fee pool explicitly so new shares do not retroactively claim past fees. That makes it an important comparator for our fee-accounting story. It also highlights the main economic difference. FPMM LP shares are claims on the pool's current inventory of conditional tokens and accrued fees. Our mechanism has to allocate a junior residual after fixed winner claims are reserved, which is why entry-time residual fairness becomes a first-class design problem here. Put differently, FPMM is operationally more permissive, while the present design is more restrictive about what late LP capital is entitled to own.

The third is the liquidity-sensitive AMM literature. Othman and Sandholm show that there is a structural tension among no-arbitrage, path independence, and liquidity sensitivity, and discuss why simply recycling fees into b is not a clean solution for LMSR-like market makers [10]. Their work is highly relevant because the active-LP problem can easily be mistaken for a generic "just increase liquidity" problem when it is really a constrained contingent-liability design problem.

The fourth is the recent bridge between prediction markets and DeFi automated market makers (AMMs). Frongillo, Papireddygari, and Waggoner show that constant-function market makers (CFMMs) and cost-function prediction markets are deeply related under a common axiomatic view [12]. Bhaskara, Frongillo, and Papireddygari then study liquidity provisioning directly for cost-function prediction markets and frame LPs as running market makers in parallel rather than merely "buying both sides" [13]. That framing is the most important direct prior art for our simulation architecture.

There are also adjacent AMM designs that attack a different part of the same problem. Paradigm's pm-AMM work studies outcome-token AMMs whose liquidity schedule shrinks as expiry approaches, precisely because prediction-market LP risk becomes extreme near resolution [11]. That is useful context for our deployment boundaries. Our skew-gate result points in the same operational direction: once a market is close to resolved, the protocol should stop pretending that additional LP capital is automatically beneficial. pm-AMM and the present paper respond to that boundary differently, but they are reacting to the same underlying pressure.

There is also a competing architectural direction based on conditional-token composability rather

than first-generation LMSR event contracts. Gnosis' conditional-tokens system was designed to improve fungibility in deeper combinatorial markets and generalizes earlier buy-all-outcomes and sell-all-outcomes operations into position splitting and merging [14, 15]. That direction is complementary rather than identical to our work: it provides a richer position graph, whereas this paper focuses on live LP entry into an LMSR-style market maker with explicit sponsor economics.

2.6 What Is New Here

The contribution is practical rather than formulaic.

- It defines a constrained live-entry LMSR mechanism with prospective-fee accounting, no active LP withdrawal, and fixed-claim winner seniority.
- It couples that mechanism to reserve-based residual release, so LP capital can unlock without requiring every winner to claim first.
- It shows, by explicit simulation against exact and implementation-oriented models, that naive residual sharing favors later LPs and that this bias can be materially reduced by a calibrated affine time-weighted rule.
- Relative to deployed pool-share systems such as FPMM, it isolates a different problem: not how to prevent new LP shares from claiming past fees, but how to allocate junior residual value fairly when winner claims remain fixed and senior.

It does not offer a theorem of optimal residual weighting. It offers a bounded sponsor-style design whose fairness bias can be measured, compared, and materially reduced.

3 Mechanism

3.1 State separation

The first architectural requirement is state separation. A live-entry mechanism cannot safely reuse a single quantity as both the pricing state and the trader-claim ledger. The market therefore has at least four conceptually distinct state domains: pricing state, trader state, sponsor state, and reserve state.

The pricing state determines current quotes and local depth. The trader state records positions and claim eligibility. The sponsor state records LP share units, fee snapshots, claimable fees, and residual entitlements. The reserve state records how much value must remain locked for unresolved winner or refund claims. This separation is not cosmetic. It is what prevents LP accounting from corrupting trader accounting.

Formally, let $n_{u,i,t}$ denote trader u's outstanding shares in outcome i at time t, and let $c_{u,i,t}$ denote the unrecovered cost basis attached to those shares. Aggregate unresolved outcome claims are

$$N_{i,t} = \sum_u n_{u,i,t}.$$

After resolution to winning outcome ω, the senior winner reserve is

$$W_t^{\text{res}} = N_{\omega,t}.$$

After cancellation, the senior refund reserve is

$$W_t^{\text{can}} = \sum_u \sum_i c_{u,i,t}.$$

The pricing state is $(B_t, \mathbf{q}_t)$, not the claim ledger. Buys, sells, winning claims, and refunds update (n, c); LP entry updates sponsor state and pricing depth; reserve release refers to W_t, not directly to $\mathbf{q}_t$. The current model assumes either that trader positions are not freely transferable during the life of the market, or that any transfer carries the remaining cost basis $c_{u,i,t}$ with the position. Without that assumption, cancellation refunds would be ill-defined or gameable.

3.2 LP entry during `ACTIVE`

An LP entering during `ACTIVE`, meaning the tradable but unresolved market state, contributes additional depth Δb. This presupposes a market that has already been bootstrapped. The mechanism is not self-starting: an initial sponsor or sponsor pool must create the first funded market state, establish the initial price vector, and supply the starting depth before later LPs can join during active trading. In the simulator and candidate mechanism, sponsor-share issuance is linear in that depth contribution:

$$s_k = \Delta b_k.$$

The entry is priced against the current market state rather than against a naive share of the existing pool. If the current price vector is $\mathbf{p}$, the required underwriting collateral is

$$D_k = \Delta b_k \cdot \alpha(\mathbf{p}), \qquad \alpha(\mathbf{p}) = \max_i \ln\left(\frac{1}{p_i}\right).$$

This is the same worst-case-loss benchmark used in the reference model's live-entry funding rule. The LP does not receive retroactive fee rights. The LP does not receive an active withdrawal path. The intended effect is to improve future slippage while preserving the current price vector up to bounded numerical tolerance.

The economic reason to issue sponsor shares in units of Δb rather than deposited collateral is that depth is the thing the LP is actually adding to the mechanism. Deposit is the underwriting price paid for that depth at the current state. Two LPs entering the same state and contributing the same depth increment should receive the same sponsor-share quantity. Denominating shares directly in deposit would instead make the sponsor unit depend on state-dependent underwriting cost rather than on depth contributed. In this design, collateral determines how much must be posted to support a chosen depth increment; it does not by itself determine the share count. That does not amount to a theorem that Δb is the uniquely correct sponsor numeraire. It is the accounting choice that best matches the economic quantity the LP is buying in the mechanism studied here.

In the exact benchmark model, each LP entry can be understood as adding a parallel LMSR component aligned to the current price vector. In the compressed protocol candidate, the same effect is represented by an update to a single aggregate pricing state. The crucial research question is whether this compression preserves the economically relevant behavior.

3.3 Why No Active Withdrawal

Disallowing LP withdrawal during `ACTIVE` is not only an implementation convenience. It is the mechanism's cleanest way to keep trader claims and sponsor claims from colliding. If capital can exit while the market maker is still carrying contingent liabilities, the protocol has to decide whose rights shrink when collateral leaves. The no-withdrawal rule removes that collision entirely.

This comes with a clear cost: capital efficiency is lower than in ordinary DeFi LP systems. The paper therefore does not describe active LPs as a generic passive-yield primitive. An LP in this design is underwriting future trading flow and junior residual value until settlement. The participation thesis is narrower. LPs should enter when they expect the combination of prospective fees and residual recovery to compensate them for lockup and junior risk. Markets where that expected return is unattractive should simply fail to attract LP capital. That is a feature of the mechanism, not a contradiction in it.

3.4 Fee rule

Fees are the cleanest part of the mechanism. A cumulative fee index tracks LP fee accrual per share unit. When an LP enters, it snapshots the current fee index. Future fees are then split among currently active LP shares. Already accrued fees are not reassigned.

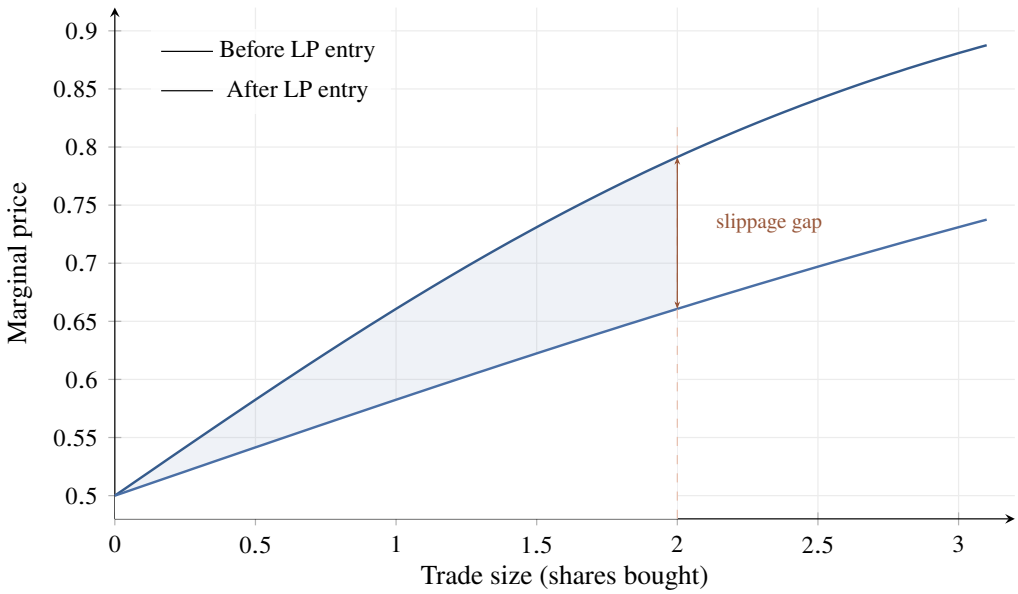

Figure 1. Didactic binary illustration (N=2, p_0=0.5). LP entry deepens the market without moving the starting price. The shaded region is the execution-cost reduction for a fixed trade size. The general multi-outcome case has the same qualitative shape.

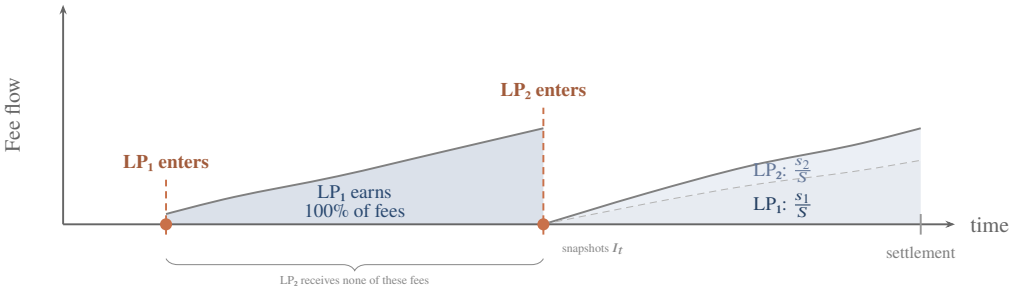

Figure 2. Prospective fee accounting. LP_1 earns all fees before LP_2 enters. After entry, fee flow is split by active sponsor shares (s_k/S_t). LP_2 snapshots the fee index I_t on entry and receives nothing from the pre-entry period.

This gives a simple and strong property: later LP entry dilutes earlier LPs only for future fees. That is the first-order fairness guarantee in the mechanism, and it is the cleanest one to preserve in implementation.

3.5 Settlement rule

The market is fixed-claim rather than parimutuel. A winning share represents a defined claim, typically one unit per winning share. This is what makes reserve-based settlement coherent. At resolution, the protocol asks how much is still owed to unclaimed winning shares. That amount is reserved. Only the remaining free pool can be considered LP residual.

The same logic applies to cancellation, except that the reserve is determined by outstanding refund liability rather than winning-share liability. In the model studied here, cancellation refunds remaining trader cost basis, not previously paid LP or protocol fees. That choice is intentional: those fees compensate liquidity service already provided and are not unwound on cancellation. This means LP capital does not need to wait for every winner to claim before any residual can be released, but LP claims can never reduce the pool below the amount still owed to winners or refund claimants.

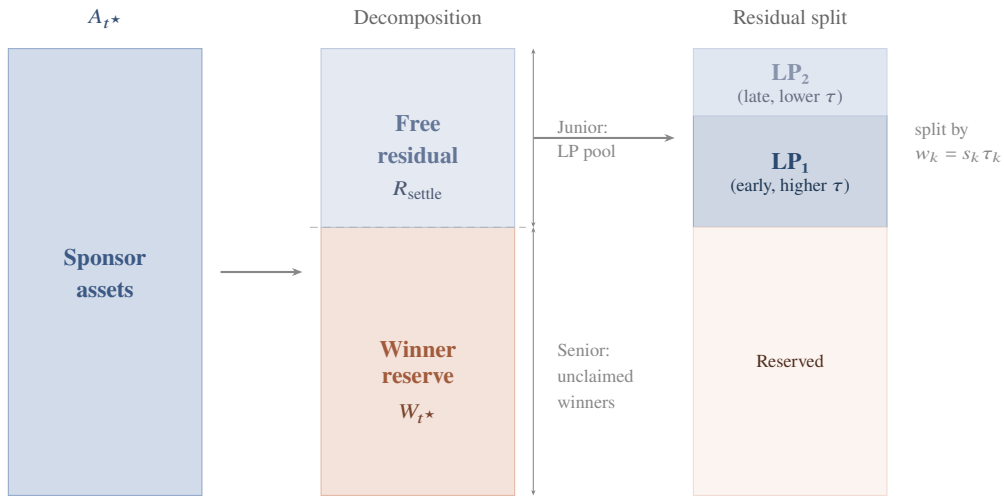

Figure 3. Reserve-based settlement. Non-fee sponsor assets ($A_{t^\star}$) are decomposed into a senior winner reserve ($W_{t^\star}$) and a junior free residual (R_{settle}). The residual is split across LP cohorts by time-weighted sponsor shares ($w_k = s_k \tau_k$).

3.6 Core transitions and reserve invariant

The mechanism is easier to judge once its central transitions are written explicitly. Let the live LMSR pricing state be $(B_t, \mathbf{q}_t)$ with price vector

$$p_i(t) = \frac{e^{q_{i,t}/B_t}}{\sum_j e^{q_{j,t}/B_t}}.$$

For a buy of x shares in outcome j by trader u, define the LMSR payment

$$\Delta C_t = C(\mathbf{q}_t + x\mathbf{e}_j) - C(\mathbf{q}_t).$$

Then the pricing and trader ledgers update as

$$q^+_{j,t} = q_{j,t} + x, \qquad n^+_{u,j,t} = n_{u,j,t} + x, \qquad c^+_{u,j,t} = c_{u,j,t} + \Delta C_t, \qquad N^+_{j,t} = N_{j,t} + x.$$

For a sell of x shares in outcome j from a trader with $0 < x \leq n_{u,j,t}$, let ΔS_t be the standard LMSR sell return. The pricing state moves by the inverse LMSR step and the claim ledger contracts proportionally:

$$n^+_{u,j,t} = n_{u,j,t} - x, \qquad c^+_{u,j,t} = c_{u,j,t}\left(1 - \frac{x}{n_{u,j,t}}\right), \qquad N^+_{j,t} = N_{j,t} - x.$$

The fee rule then decomposes trader payment or return into quote cost, LP fee flow, and protocol fee flow, with the LP fee portion passed into the cumulative fee index already defined in §2.2. After resolution to winning outcome ω, winner claims burn $n_{u,\omega,t}$ and reduce W^{res}_t one-for-one. After cancellation, refunds reduce $c_{u,i,t}$ and W^{can}_t one-for-one.

For active LP entry, let the pre-entry state be $(B_t, \mathbf{q}_t, \mathbf{p}_t)$ and let the entrant add depth $\Delta b > 0$. The aligned entry component is

$$u_i(\Delta b, \mathbf{p}_t) = \Delta b\,(\ln p_i(t) + \alpha(\mathbf{p}_t)), \qquad \alpha(\mathbf{p}_t) = \max_j \ln\left(\frac{1}{p_j(t)}\right).$$

The compressed Layer B transform is then

$$B_t^+ = B_t + \Delta b, \qquad q_{i,t}^+ = q_{i,t} + u_i(\Delta b, \mathbf{p}_t).$$

This preserves prices by construction up to gauge. Because any LMSR state can be written in the form $q_{i,t} = B_t(\ln p_i(t) + c_t)$ for some common shift c_t, and the new component is aligned to the same $\mathbf{p}_t$, there exists another common shift c_t^+ such that

$$q_{i,t}^+ = (B_t + \Delta b)\left(\ln p_i(t) + c_t^+\right),$$

which implies $\mathbf{p}_t^+ = \mathbf{p}_t$. The same transform also weakly improves local slippage for fixed same-direction probe trades, because the post-entry depth parameter is larger while the pre-trade price vector is unchanged.

Layer A makes the same economics explicit in parallel form. Let component m have depth $b_t^{(m)}$ and pricing state $\mathbf{q}_t^{(m)}$, with aggregate

$$B_t = \sum_m b_t^{(m)}, \qquad \mathbf{q}_t = \sum_m \mathbf{q}_t^{(m)}.$$

For a trade of size x in outcome j, the reference router splits the trade in proportion to component depth:

$$x^{(m)} = x \cdot \frac{b_t^{(m)}}{B_t}, \qquad \sum_m x^{(m)} = x.$$

Each component then updates by the standard LMSR rule

$$q_{j,t}^{(m)+} = q_{j,t}^{(m)} + x^{(m)},$$

with all other coordinates unchanged. The same proportional router is used for sells. If the components are aligned before the trade, meaning each can be written as

$$q_{i,t}^{(m)} = b_t^{(m)}(\ln p_i(t) + c_t^{(m)}),$$

then they remain aligned after routed trading, and the aggregate state evolves exactly like a single LMSR with depth B_t. This algebraic identity explains the Layer A to Layer B match: the compressed implementation preserves the same update rule in high precision.

The cash side aggregates exactly as well. For a routed buy in outcome j, each aligned component charges

$$\Delta C_t^{(m)} = b_t^{(m)} \ln\left(1 - p_j(t) + p_j(t)e^{x^{(m)}/b_t^{(m)}}\right).$$

Because $x^{(m)}/b_t^{(m)} = x/B_t$ under the proportional router, this becomes

$$\Delta C_t^{(m)} = \frac{b_t^{(m)}}{B_t} B_t \ln\left(1 - p_j(t) + p_j(t)e^{x/B_t}\right) = \frac{b_t^{(m)}}{B_t}\Delta C_t.$$

Summing across components gives

$$\sum_m \Delta C_t^{(m)} = \Delta C_t.$$

So not only the aggregate pricing state but also the aggregate trade-cash update, and therefore sponsor NAV accounting, coincide exactly between Layer A and Layer B whenever the components remain aligned.

The local-slippage claim can also be stated more sharply. For a fixed pre-trade outcome price p_j and a buy of $x > 0$ shares in that outcome, the LMSR buy cost can be written as

$$f_B(x; p_j) = B \ln\left(1 - p_j + p_j e^{x/B}\right).$$

Let $h(y) = \ln(1 - p_j + p_j e^y)$ with $y = x/B$. Then

$$\frac{\partial f_B}{\partial B} = h(y) - y h'(y).$$

Since h is convex and $h(0) = 0$, we have $h(y) \leq y h'(y)$ for $y \geq 0$, which implies

$$\frac{\partial f_B}{\partial B} \leq 0.$$

So at a fixed pre-trade price, increasing depth weakly lowers the buy cost for the same probe trade. That is the analytic reason the slippage checks in the simulator pass after price-preserving LP entry.

Reserve-based settlement can also be stated as a simple accounting invariant. Let A_t denote non-fee sponsor assets net of previously claimed LP residual, and let W_t denote the outstanding winner or refund reserve. Define the free residual pool as

$$L_t = \max(0, A_t - W_t).$$

If K_t denotes total contract funds, P_t protocol-fee liabilities, and G_t LP-fee liabilities, then the paper's non-fee sponsor-asset quantity is simply

$$A_t = K_t - P_t - G_t.$$

Winner or refund claims of size x reduce both sides of the senior book:

$$A_t^+ = A_t - x, \qquad W_t^+ = W_t - x,$$

so the free residual pool remains unchanged:

$$L_t^+ = \max(0, A_t^+ - W_t^+) = \max(0, A_t - W_t) = L_t.$$

LP-fee withdrawals are separate again:

$$K_t^+ = K_t - z, \qquad G_t^+ = G_t - z, \qquad A_t^+ = A_t, \qquad W_t^+ = W_t.$$

So fee withdrawals reduce the fee-liability bucket without invading winner reserve collateral.

To make LP claim ordering well-defined, the weighted residual split is snapshotted at settlement rather than recomputed from the shrinking post-claim pool. Let $t^\star$ denote the first settled state before any LP residual claims, and define

$$R_{\text{settle}} = \max(0, A_{t^\star} - W_{t^\star}).$$

Let the settlement-share weights be

$$\pi_k = \frac{w_k}{\sum_j w_j}, \qquad \bar{R}_k = \pi_k R_{\text{settle}}.$$

Let $C_{k,t}$ denote residual already claimed by cohort k after settlement. Then the remaining cohort entitlement is

$$E_{k,t} = \bar{R}_k - C_{k,t},$$

and the remaining free residual pool is

$$L_t = R_{\text{settle}} - \sum_k C_{k,t} = \sum_k E_{k,t}.$$

An LP residual claim by cohort k of size y_k is permitted only if $0 \le y_k \le E_{k,t}$, in which case

$$A_t^+ = A_t - y_k, \qquad W_t^+ = W_t, \qquad C_{k,t}^+ = C_{k,t} + y_k, \qquad E_{k,t}^+ = E_{k,t} - y_k, \qquad L_t^+ = L_t - y_k \ge 0.$$

This is the clean reason two different properties hold at once. Winner seniority is protected because LP claims can never reduce the pool below W_t. Claim-order invariance also holds because the final cohort allocation is determined by the settlement snapshot $\bar{R}_k$, not by the order in which different cohorts happen to withdraw from the shrinking free pool.

3.7 Residual weighting

Once reserve-based release is adopted, a second question remains: how should the free residual pool be split across LP cohorts? This is where the economics become genuinely interesting. A naive unweighted sponsor pool favors later entrants because they buy into a market after some uncertainty has already been resolved. A fully linear time-weight rule goes too far in the opposite direction by strongly favoring earlier LPs.

The simulator's current affine rule is explicit. Let t_k be cohort k's entry time, let T be the settlement timestamp, and define elapsed exposure

$$e_k = \max(1, T - t_k).$$

The cohort's time weight is then

$$\tau_k^{\text{step}} = 1 + \lambda_{\text{step}}(e_k - 1),$$

which is the event-step rule used in the paper's first-stage calibration sweeps. The protocol-facing rule studied later normalizes that exposure by market lifetime:

$$a_k = \begin{cases} 0, & T - t_0 \le 1, \\ \dfrac{e_k - 1}{T - t_0 - 1}, & T - t_0 > 1, \end{cases}$$

and its time weight is

$$\tau_k^{\text{norm}} = 1 + \lambda_{\text{norm}} a_k.$$

The corresponding residual weight is

$$w_k = s_k \tau_k.$$

If the free residual pool at settlement is R, the cohort's pre-claim entitlement is

$$R_k = R \cdot \frac{w_k}{\sum_j w_j}.$$

The current research result is that a mild affine time premium performs materially better than both flat pooling and full linear weighting, but the clock used to measure exposure matters. A first-stage event-step sweep found a near-neutral region around $\lambda_{\text{step}} = 0.03250$ on the midpoint benchmark. A stronger held-out study then restamped the same logical paths across short, medium, and long scenario clocks and selected a lifetime-normalized premium $\lambda_{\text{norm}} = 0.150$. That normalized rule is the current paper and protocol default because it generalizes much better across duration-scaled held-out packs. Both parameters are empirical and provisional, not canonical. What matters is not the constant by itself. What matters is that the bias has direction and magnitude, so it can be measured, compared, and calibrated.

Event-time exposure is also only a proxy for the more complicated thing the mechanism is really trying to capture: how much uncertainty and liability evolution a cohort has absorbed by entering earlier rather than later. It is not a claim that elapsed time is the unique sufficient statistic for residual fairness. Trade flow, information flow, volatility, or liability-path exposure are also plausible bases for weighting. The present paper still uses event-indexed timestamps because they are transparent, cheap to compute, and easy to carry into a protocol, but it normalizes those timestamps by market lifetime before turning them into the default residual premium.

3.8 Worked example

The mechanism is easiest to internalize with one stylized example. The purpose of this example is to make the accounting concrete, not to reproduce the full average fairness result of the simulation pack in one toy path. Consider a five-outcome market with one bootstrap sponsor and one late entrant.

- The bootstrap sponsor enters first at the uniform start state $\mathbf{p}_0 = (0.2, 0.2, 0.2, 0.2, 0.2)$. Then $\alpha(\mathbf{p}_0) = \ln 5 \approx 1.609$, so with share units $s_0 = 100$ the required collateral is $D_0 \approx 160.9$.
- Before any other LP arrives, trading generates 4 units of LP fees. Those fees belong entirely to the bootstrap sponsor.
- The market then moves to a moderately skewed state $\mathbf{p}_1 = (0.40, 0.20, 0.15, 0.15, 0.10)$. Here $\alpha(\mathbf{p}_1) = \ln 10 \approx 2.303$, so a late LP entering with share units $s_1 = 50$ posts $D_1 \approx 115.1$ of collateral. That late LP receives none of the earlier 4 fee units.
- After entry, trading generates 6 more fee units. Those are split by active share units, so the bootstrap sponsor receives 4 and the late LP receives 2.
- At settlement, after winner claims are reserved, the free residual pool is 300.

Under flat residual pooling, the residual weights are just 100 and 50, so the residual split is 200 and 100. Under an affine rule with time weights 1.30 for the early cohort and 1.00 for the late cohort, the residual weights become 130 and 50, so the residual split is approximately 216.7 and 83.3. The opposite sign, which is the one that motivates the paper's calibration program, is just as easy to see in a one-line micro-example. Suppose two equal-size cohorts each ultimately receive half of a flat residual pool of 200, but the earlier cohort entered when the underwriting floor was $\alpha = 2.303$ and the later cohort entered when it was $\alpha = 1.609$. With $s_0 = s_1 = 50$, the deposits are then approximately 115.1 and 80.5, so the residual-only NAV-per-deposit ratios are about 0.869 and 1.242. Flat residual pooling therefore gives the later entrant a materially better capital-normalized outcome even though both cohorts ultimately split the same pooled residual. The affine rule exists to reduce that kind of timing subsidy, not to rewrite fee fairness.

Table 1. Mechanism summary.

Component	Rule	Rationale
Initial market funding	Required before `ACTIVE`	Establishes the first price vector and starting depth.
LP entry	Allowed during `ACTIVE`	Lets depth grow after launch.
LP withdrawal	Disallowed during `ACTIVE`	Avoids impairing trader rights and winner claims.
Fee rule	Prospective only	Late LPs do not receive retroactive fees.
Winner payout model	Fixed-claim, senior	Makes reserve-based settlement coherent.
Residual release	Reserve-based	Unlocks LP capital without requiring all winners to claim first.
Residual weighting	Mild affine time premium	Reduces systematic bias in pooled residual distribution.
High-skew late entry	Likely gated in production	Near-resolved markets punish late LPs rather than rewarding them.
Current provisional default	$\lambda_{\text{norm}} = 0.150$	Held-out near-neutral with low duration drift.

Figure 4. Three-layer simulation architecture. Layer A defines the exact economic benchmark. Layer B compresses to a single global state suitable for on-chain implementation. Layer C tests that design under fixed-point arithmetic and AVM constraints.

Cohort	Flat NAV / deposit	Affine NAV / deposit
Bootstrap LP	$(4+4+200)/160.9 \approx 1.293$	$(4+4+216.7)/160.9 \approx 1.397$
Late LP	$(2+100)/115.1 \approx 0.886$	$(2+83.3)/115.1 \approx 0.741$

The numbers are illustrative rather than taken from a single simulation path, but the mechanics are exact. Fee fairness is strict. The late LP receives no retroactive fees. The correction happens in the residual split, not in the fee rule. In this particular toy path, affine weighting moves additional residual value toward the earlier cohort because it carried the market through more of the active period. The point of the examples is to show where the rule acts and how fee fairness and residual fairness are separated.

4 Simulation Program

4.1 Layer A: exact reference model

Layer A is the mathematical benchmark. It represents active LP entry as parallel LMSR components in high-precision arithmetic. Each component has its own collateral basis and sponsor accounting

context, and trades are routed by an explicit rule rather than by an implicit approximation. Layer A is not intended to be protocol-ready. Its purpose is to define the clean economic target.

This matters because later layers should be judged against it. If a compressed implementation differs from Layer A in high precision where no approximation is conceptually required, that is a design problem rather than an acceptable implementation concession. At the same time, Layer A is still a modeling choice. It is the paper's normative reference model for sponsor-side economics, not a theorem that all reasonable decompositions must coincide with it.

4.2 Layer B: compressed protocol candidate

Layer B represents the likely on-chain mechanism: one global pricing state, one aggregate depth parameter, a cumulative fee index for prospective fees, and sponsor accounting consistent with reserve-based settlement. The key question for Layer B is whether a compressed global-state implementation can preserve the relevant economics of Layer A closely enough to justify a real protocol implementation.

4.3 Layer C: fixed-point / AVM-oriented model

Layer C takes the Layer B economics and pushes them through a fixed-point approximation shaped by the Algorand Virtual Machine (AVM), the execution environment question.market uses on-chain. This focus is specific to the product, so it deserves a short justification. question.market is built on Algorand because the chain combines instant finality, negligible transaction costs, and smart-contract features strong enough to make an LMSR-based market practical. The same choice also imposes concrete engineering constraints: fixed-point arithmetic, bounded storage, and opcode-budget limits. Layer C exists to make those constraints visible early. A design that is elegant in Decimal arithmetic can still fail once it is translated into the arithmetic and resource limits of the target chain. Layer C is therefore not an afterthought; it is the boundary between research plausibility and protocol plausibility for an Algorand deployment.

The current Algopy market contract already follows this AVM-shaped decomposition. In the generated ARC-56 schema, the app uses 39 global uint64 slots, 13 global byte slots, and 5 local uint64 slots, while dynamic ledgers are pushed into a small box layout: per-outcome quantities, a packed total-user-shares box, and per-user boxes for outcome shares, claimable fees, cost basis, and pending payouts. The hot lifecycle is also split across separate ABI methods — bootstrap, buy, sell, active LP entry, LP fee settlement, residual claim, and finalization — rather than one monolithic call. This is concrete implementation evidence that the Layer C design maps onto real AVM storage primitives and call boundaries, even though exact per-method opcode and box-reference profiling still belongs to deployment hardening rather than to the present simulation study.

4.4 Scenarios and Evaluation

The simulator uses three scenario families.

Deterministic scenarios are used for sanity checks and sharp mechanism tests such as zero-flow NAV invariance, same-block reordering, and reserve claim ordering. Monte Carlo scenarios explore broader state space under randomized but structured market paths. Adversarial scenarios intentionally search for tails, especially those involving late entry into skewed markets, reserve-order effects, and low-tail many-outcome stress.

Every major experiment family is judged along a small set of explicit checks and diagnostics: price continuity, slippage improvement, LP fairness by entry time, aggregate reserve coverage, cohort sponsor-solvency diagnostics, required order invariance, and divergence between exact and simplified models. Price continuity and reserve coverage are treated as hard invariants. Slippage improvement is reported as a probe metric on fixed buy-side trades after LP entry, because the question at that stage is whether new depth makes subsequent demand-side execution cheaper without changing price.

The scenario generators are structured rather than open-ended. Deterministic bundles are hand-built

cases such as neutral late entry, skewed late entry, repeated LP entries, reserve claim ordering, and cancellation/refund paths. Ordinary Monte Carlo bundles sample 3, 5, or 8 outcomes; initial depth 80, 100, or 140; LP fees of 50, 100, or 200 basis points; protocol fees of 0 or 25 basis points; 4–8 trade events; 1–3 LP entries placed uniformly across event slots; buy sizes from $\{2, 4, 6, 8\}$ shares; sell probability 0.2; and cancel probability 0.15, with the resolved winner sampled from the terminal price vector. The adversarial bundles use 3 or 5 outcomes, fixed initial depth 100, late-entry depth increments of 20 or 50, pre-entry one-sided flow of 6 or 20 shares, optional post-entry trend or reversion trades of 12 shares, counterflow ratios of 0 or 0.25, and favorite-versus-hedge resolution policies. The dedicated high-skew boundary packs increase the outcome count to 16, 32, or 64 and use one or two 50–500 share one-sided trades with a single late entry to drive entry-time maximum probabilities above 0.8.

The primary metrics are defined as follows.

- **Price continuity:** the maximum absolute componentwise change in the price vector immediately before and after LP entry.
- **Slippage pass:** whether a fixed probe set of post-entry buy quotes is weakly cheaper than the corresponding pre-entry quotes.
- **LP fairness by entry time:** the difference in NAV per deposited unit between the latest-entering and earliest-entering LP cohorts in a scenario. For cohort k at the evaluation endpoint t^*, we define
$$V_k(t^*) = R_k^{\text{mtm}}(t^*) + F_k^{\text{claimable}}(t^*) + F_k^{\text{withdrawable}}(t^*),$$
where the residual term is marked to the current price vector in active states and equals the cohort's currently claimable reserve-pool share in resolved or cancelled states.
- **Aggregate reserve coverage:** whether non-fee sponsor assets remain at least as large as the outstanding winner or refund reserve. In reserve-policy tables, this is the quantity reported as the pass-rate row.
- **Cohort sponsor-solvency:** whether every cohort's attributed residual basis remains nonnegative across outcomes. In the exact benchmark this diagnostic is
$$M_{k,i,t} = D_{k,t}^{\text{locked}} + X_{k,t}^{\text{trade}} - \left(q_{k,i,t}^{\text{cur}} - q_{k,i}^{\text{base}}\right),$$
where $D_{k,t}^{\text{locked}}$ is cohort k's remaining locked collateral, $X_{k,t}^{\text{trade}}$ is its accumulated trade-cash balance, $q_{k,i,t}^{\text{cur}}$ is its current component state in outcome branch i, and $q_{k,i}^{\text{base}}$ is its entry baseline. The check passes when $\min_{k,i,t} M_{k,i,t} \geq 0$. A negative value is treated as a cohort-accounting pathology and deployment boundary, not as a failure of winner seniority or aggregate reserve coverage.
- **Validation or diagnostic failure:** any named validation check in the scenario evaluation payload that returns `passed = false`.
- **Required order invariance:** divergence in terminal state across alternative event orderings that are intended to be accounting-equivalent, such as post-settlement claim orderings or scenario-bundle evaluation orderings. Economically distinct sequences, such as trade-before-LP-entry versus LP-entry-before-trade, are not expected to agree and are not treated as violations.

The reported slippage pass rate is a probe statistic rather than a named invariant in the current evaluation payload. This is why a pack can show slippage pass rate slightly below one without increasing the invariant-failure count.

For Layer C specifically, the paper uses two regime labels. The ordinary regime is the target fixed-point pack with 3, 5, or 8 outcomes, moderate depth, and ordinary trading activity. The low-tail regime is the stress pack with 16, 32, or 64 outcomes and one or two large one-sided trades that drive some active probabilities close to zero.

Table 2. Midpoint calibration benchmark by scenario family.

Family	Count	Outcomes	Trade events	Construction
Deterministic	10	fixed	fixed	Hand-built sanity and edge cases: neutral and skewed late entry, repeated LP entry, reserve claim ordering, and cancellation/refund paths.
Monte Carlo	64	3, 5, 8	4–8	Initial depth 80/100/140, 1–3 LP entries, buy sizes 2/4/6/8, sell probability 0.2, cancel probability 0.15, winner sampled from terminal prices.
Adversarial	96	3, 5	1–3 core phases	Fixed depth 100, pre-entry skew of 6 or 20 shares, late-entry depth 20 or 50, optional post-entry trend/reversion trade of 12 shares, favorite-versus-hedge resolution.

4.5 Calibration Workload

The residual-weight parameter is calibrated on a mixed workload rather than on a single headline scenario. The original midpoint benchmark contains ten deterministic scenario bundles, sixty-four Monte Carlo bundles, and ninety-six adversarial bundles. That staged search is still useful because it first shows that the bias has a narrow crossover at all. But it is no longer the end of the story.

The stronger calibration program used in this draft is train/test and duration-aware. Deterministic bundles still run as always-on mechanical constraints. The ordinary-regime calibration objective is then evaluated on sixty-four Monte Carlo train bundles and sixty-four held-out test bundles, each restamped onto short, medium, and long scenario clocks of 12, 48, and 168 steps, plus ninety-six ordinary adversarial train bundles and ninety-six held-out adversarial bundles split by enumeration parity. Both the event-step family and the normalized-exposure family are scored on the same train/test structure.

The objective is deliberately narrow: choose the parameter that minimizes the absolute mean late-minus-early NAV-per-deposit gap on train while preserving price continuity, slippage, and aggregate reserve coverage. NAV is normalized by deposit because cohorts enter with different capital commitments and the paper is trying to compare economic treatment on a common unit basis. The early-versus-late axis is primary because timing advantage is the specific asymmetry created by live entry. This fairness target is an ex post realized diagnostic used as a proxy for timing subsidy, not a theorem-derived social-welfare optimum or a claim that the mechanism equalizes risk-adjusted returns.

The resulting comparison is sharp. The best event-step train winner is $\lambda_{\text{step}} = 0.02500$, but on held-out ordinary-regime packs it still posts mean gap about -0.1847, mean absolute gap about 0.2131, and duration-bucket max absolute mean gap about 0.5301. The best normalized train winner is $\lambda_{\text{norm}} = 0.150$, and on held-out ordinary-regime packs it posts mean gap about -0.0068, mean absolute gap about 0.0394, and duration-bucket max absolute mean gap about 0.0223. That is why the normalized rule becomes the default in the paper rather than remaining a future-work suggestion.

The calibration should also not be confused with an equilibrium solution. The adversarial bundles in the current paper are strategically constructed timing scenarios evaluated under a fixed published rule. They are useful because they expose the sign and magnitude of timing advantage under stress.

Table 3. Held-out calibration comparison. The normalized family generalizes far better across duration-scaled ordinary-regime packs than the best event-step family.

Mode	Selected parameter	Held-out mean gap	Held-out mean absolute gap	Held-out duration drift
Event-step affine	$\lambda_{\text{step}} = 0.02500$	−0.1847	0.2131	0.5301
Normalized affine	$\lambda_{\text{norm}} = 0.150$	−0.0068	0.0394	0.0223

They are not a proof that fully adaptive agents who know the published weighting rule and optimize entry timing against it have no remaining edge. The current evidence supports a bias-reduced default, not strategy-proofness against rule-aware timing arbitrage. It is also still a held-out split inside one scenario-generator family rather than a theorem of universal portability.

The ordinary calibration packs also have a finite skew range. In the held-out normalized test rows, the largest entry-time maximum outcome probability is about 0.3792. The default should therefore be read as calibrated for ordinary and moderately skewed late entry, not for capital arriving when one outcome is already near-certain.

4.6 Adversarial timing model

The adversarial scenarios in this paper are not meant to be a full strategic equilibrium, but they do correspond to a concrete threat model. The adversary observes the public market state, including the current price vector, active depth, active sponsor shares, and published residual-weight rule. It then chooses an entry time, a depth increment from the tested schedule, and, in some scenario families, a placement relative to skewing or reversionary trades. The evaluation objective is simple: maximize the late-entry advantage measured on the same NAV-per-deposit fairness metric used throughout the paper. This is a meaningful stress model for deployment because it asks whether a rule-aware LP can still improve its economics by timing entry against known settlement and residual rules. It is not a theorem of optimal adversarial play, and it does not cover the full joint strategy space of LPs and traders.

Nor does the current adversarial model include trader sequencing around a publicly visible LP-entry transaction. Because LP entry is price-continuous by construction, the concern there is not free repricing arbitrage at the entry event itself, but ordering advantage around the cheaper post-entry execution path once additional depth is known to be arriving.

4.7 Validation philosophy

The simulator is evidence, not a proof assistant. For that reason, this paper separates validated claims from provisional claims and deployment boundaries. A mechanism can be strong enough to support implementation even when some hard boundary conditions still need to be enforced operationally. The standard here is traceable evidence, honest disclosure, and a clearly defined deployment envelope.

4.8 FPMM comparator

Because the closest deployed active-LP comparator is the Gnosis fixed-product market maker, the simulation program also includes a matched resolved-only head-to-head against an FPMM-style pool-share baseline [8, 9]. The comparison holds fee rates, outcome count, trader event sequence, LP entry timing, and LP capital schedules fixed, while allowing the quote function, pool-share minting rule, and settlement accounting to remain mechanism-specific. In the comparator pack, FPMM LPs are held through settlement even though the production contract supports withdrawal. That choice isolates the entry-timing fairness question rather than mixing it with a separate withdrawal-timing game. It should therefore be read as a controlled diagnostic benchmark rather than as a full native-strategy comparison of FPMM behavior.

Table 4. Layer B matches the chosen reference model in high precision.

Metric	Value
Result count	1682
Price continuity pass rate	1
Slippage pass rate	1
Reserve-coverage pass rate	1
Mean scenario max quote divergence	4.4009×10^{-27}
Max scenario max quote divergence	3.2738×10^{-26}
Mean scenario max NAV/deposit divergence	8.0143×10^{-29}
Max scenario max NAV/deposit divergence	1.0×10^{-27}
Validation / diagnostic failure count	0

The comparator mechanics are explicit. The bootstrap pool starts with equal balances across outcomes. If a later FPMM entrant deposits collateral D into a skewed pool with balances B_i and current LP share supply S, the benchmark follows the Gnosis-style proportional minting rule against the pool's largest balance $P = \max_i B_i$:

$$r_i = D \cdot \frac{B_i}{P}, \qquad e_i = D - r_i, \qquad m = D \cdot \frac{S}{P}.$$

Here r_i is the amount of outcome-i inventory actually added to the pool, e_i is the excess inventory returned to the entrant outside the pool, and m is the number of pool shares minted. Comparator NAV includes four pieces: the entrant's pro rata claim on terminal pool inventory, the marked value of returned external outcome tokens, accrued claimable fees, and withdrawable fees. Transfers are not modeled, and active withdrawal is disabled only to keep the comparison focused on entry-timing economics rather than withdrawal timing.

This comparator should therefore be read as a comparison of timing-sensitive LP economics, not as a claim that one mechanism dominates the other in all respects. FPMM largely sidesteps the residual-weighting problem because LP equity is expressed as pool inventory and accrued fees. That does not make timing irrelevant. It moves the timing risk into the composition of the pool inventory itself.

5 Results

5.1 Layer B Matches the Reference Model

The clearest positive result is that the compressed Layer B model tracks the chosen reference model extraordinarily well in Decimal arithmetic. In the core comparison pack, the mean maximum quote divergence versus the reference benchmark is approximately 4.40×10^{-27} and the mean maximum NAV-per-deposit divergence is approximately 8.01×10^{-29}, with price continuity, slippage improvement, and aggregate reserve coverage all passing across 1,682 results. In practical terms, this means the main implementation compression appears to preserve the economics of the reference model rather than merely approximate them loosely.

This is a crucial result because it converts the active-LP model from a mathematically interesting construction into something that plausibly fits a protocol architecture. The claim here is benchmark-relative. It does not, by itself, prove that no other sponsor-accounting interpretation is possible.

5.2 Reserve Release Solves Settlement but Favors Late LPs

The reserve-based residual baseline solves one operational problem cleanly: it avoids the impractical requirement that every winner must claim before LP capital can unlock. This is stronger than a simulation anecdote. Under the reserve invariant in §3.6, winner claims reduce both the pool and the reserved senior liability by the same amount, while LP residual claims are bounded above by the free

pool $L_t = \max(0, A_t - W_t)$. As a result, LP claim ordering cannot invade winner seniority under the modeled rule. The reserve-mode scenarios are useful because they confirm that this accounting logic survives the full state machine, not because the claim itself depends on Monte Carlo luck.

However, the unweighted reserve pool is not economically neutral. It systematically favors later entrants. In the quick reserve pack, the mean fairness gap between late and early LPs is approximately +0.1074, with maximum absolute gap about 0.2221. In plain terms, naive residual sharing rewards waiting. Residual weighting is necessary if the mechanism wants to reduce that subsidy.

5.3 Late-LP Advantage Is Calibratable

The calibration experiments show a useful structure rather than a dead end. Fully linear time-weighting strongly overcorrects toward earlier LPs, with mean fairness gap about −0.6661. Mild affine weighting performs much better. The important refinement is that the paper now treats the old event-step midpoint result as a first-stage discovery rather than as the final answer. The midpoint sweep around $\lambda_{\text{step}} = 0.03250$ showed that neutrality was reachable on the original mixed pack. The held-out duration-aware study then showed that the event-step family itself is too clock-sensitive to keep as the lead default.

That is why the current default is the normalized rule rather than the earlier midpoint. With $\lambda_{\text{norm}} = 0.150$, the held-out ordinary-regime pack yields mean late-minus-early NAV-per-deposit gap about −0.0068, mean absolute gap about 0.0394, median gap about −0.0131, 5th percentile about −0.0737, and 95th percentile about +0.0828, with price continuity, slippage, and aggregate reserve coverage all passing and no validation or diagnostic failures. More importantly, the short-, medium-, and long-duration Monte Carlo buckets cluster tightly around zero, with means approximately −0.0003, +0.0014, and +0.0018. On the same held-out structure, the best event-step winner still drifts from about −0.0487 in the short bucket to about −0.5301 in the long bucket. That is not a small calibration nuisance. It is the empirical reason the default should be expressed on a normalized lifetime clock.

It is also important not to overstate what "near neutrality" means. Even the normalized rule is not minimax or strategy-proof. In the held-out adversarial slice of the selected normalized pack, the mean late-minus-early gap is approximately −0.0223 and the mean absolute gap about 0.0341, with worst observed negative gap about −0.0825 and worst positive gap about +0.0614. The rule therefore reduces timing bias sharply on the intended workload and removes the systematic late-entry premium in the held-out adversarial slice, but strategic timing remains economically meaningful.

Near-resolved entry is a separate boundary case, not something the ordinary calibration packs quietly solve. In a dedicated normalized high-skew boundary pack, once the maximum entry-time outcome probability exceeds 0.8, the mean late-minus-early NAV-per-deposit gap turns negative at approximately −0.2445. Above 0.9, it falls further to approximately −0.3258. This is more than a story about limited upside. In that regime the calibrated rule has flipped sign and late entry becomes a systematically unfavorable trade. A late LP entering a market that is already close to resolved is often posting collateral into a market with too little residual upside left to justify junior exposure. The affine rule helps balance ordinary workloads. It does not make near-resolved late entry universally attractive, and it should not be expected to.

The calibration path is not merely described in prose. It is captured in the public artifact set associated with this paper, including the calibration figure, comparison figure, and machine-readable summary tables published in the active-LP repository.

5.4 Compared with FPMM, the main difference shows up at the boundary

The resolved-only FPMM comparison helps sharpen what this paper is and is not claiming. The refreshed comparator now uses the same normalized default selected by the held-out calibration program, so the comparison no longer depends on the older event-step parameterization. FPMM still does not need a special residual-weighting rule because LP shares are claims on pool inventory, returned external outcome tokens, and accrued fees. It is, in that sense, the more operationally permissive mech-

Table 5. Residual rules. Naive pooling favors later LPs, linear time weighting favors earlier LPs, the old event-step midpoint shows the first in-sample crossover, and the normalized held-out default becomes the protocol-facing rule.

Rule	Mean fairness gap	Max absolute fairness gap	Validation / diagnostic failures	Interpretation
Flat reserve	+0.1074	0.2221	0	Late-LP favoring baseline.
Linear time-weighted	−0.6661	1.3645	0	Overcorrects toward early LPs.
Affine event-step midpoint ($\lambda_{\text{step}} = 0.03250$)	+0.00012	0.2075	0	Useful in-sample crossover, but too clock-sensitive to keep as the default.
Affine normalized held-out ($\lambda_{\text{norm}} = 0.150$)	−0.0068	0.1394	0	Current paper and protocol default.

anism: LPs can add funding, remove funding, withdraw fees, and transfer their pool shares. On some ordinary and moderately adversarial paths, that pool-share model remains competitive on the same late-minus-early NAV-per-deposit metric used throughout this paper. In the adversarial comparator slice, for example, the FPMM mean fairness gap is approximately +0.0173 versus about +0.0109 for the normalized active-LP mechanism, and the active-LP design has the smaller absolute gap in fifteen of twenty-four tested bundles.

The picture changes even more sharply in the high-skew boundary regime. There the normalized active-LP comparator produces a mean fairness gap of about −0.1203, while the FPMM baseline rises to about +0.5373. Across all twelve high-skew paired scenarios in the comparator pack, the active-LP mechanism has the smaller absolute fairness gap. The practical interpretation is straightforward. FPMM avoids the junior-residual accounting problem, but it does not avoid timing-sensitive LP economics. In highly skewed markets, the timing problem appears through pool-inventory composition rather than residual allocation, and the late-entry advantage can become much larger. On the evidence in this paper, the pool-share inventory model appears more exposed in the tested high-skew comparator boundary.

Taken across the full resolved comparator pack, the normalized active-LP mechanism has mean fairness gap about −0.0243, compared with about +0.1615 for the FPMM baseline, and it has the smaller absolute fairness gap in twenty-nine of forty-four paired scenarios. That is not a universal dominance result. The narrower claim is that, in the tested high-skew deployment boundary, the active-LP design compares favorably to the closest deployed pool-share alternative.

5.5 Layer C Is Strong in Ordinary Regimes, but Low-Tail Is a Real Boundary

The ordinary Layer C regime is encouraging. Price continuity and aggregate reserve coverage pass. Mean quote divergence versus the reference benchmark is about 0.0001669, and mean NAV-per-deposit divergence is about 0.0004717. These values are small enough to suggest the mechanism can survive a realistic fixed-point environment in normal states. Layer C uses the same economic transitions as Layer B but with uint64 fixed-point prices and state variables at scale 10^6, 20-term

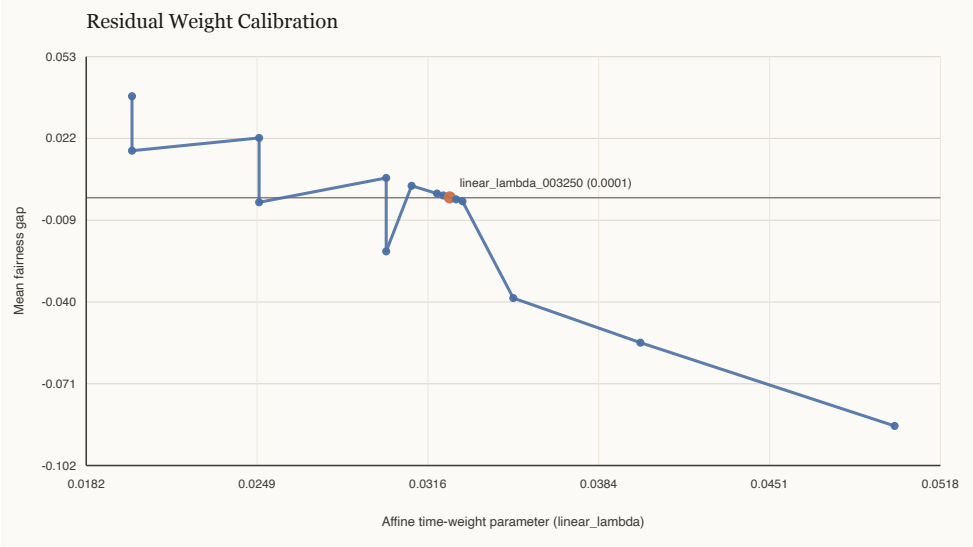

Figure 5. First-stage residual-weight calibration. Unweighted residuals favor later LPs, fully linear weighting overcorrects, and the event-step affine family crosses near neutrality near lambda = 0.03250 before the held-out normalized upgrade.

Table 6. Active-LP versus FPMM on matched resolved scenario families. Lower absolute fairness gap is better. The difference column reports signed mean-gap arithmetic, not absolute-gap arithmetic.

Slice	Count	Active-LP gap	FPMM gap	Signed mean gap diff	Active wins	FPMM wins
Deterministic	8	+0.0141	+0.0305	+0.0165	2	6
Adversarial	24	+0.0109	+0.0173	+0.0065	15	9
High-skew	12	−0.1203	+0.5373	+0.6576	12	0
Overall	44	−0.0243	+0.1615	+0.1859	29	15

exponential and 32-term logarithm approximations, floor rounding for prices and routed share splits, ceiling rounding for user-facing costs and collateral, a one-unit price floor, and a one-unit entry safety margin. That is enough to support implementation within a documented operating envelope.

The low-tail regime is the main stress boundary. In many-outcome, highly skewed states, mean quote divergence rises to about 0.0199, maximum quote divergence reaches about 0.2726, and invariant failures appear. The important distinction is that aggregate reserve coverage still passes in this pack: winner seniority is not the thing that breaks. All eight validation / diagnostic trips are instead `sponsor_solvency` failures, spread across three Monte Carlo paths in the 16-, 32-, and 64-outcome slices, and they appear in both the exact reference model and the AVM-oriented model. This means the low-tail issue is not merely an implementation artifact. It is a mechanism boundary in extreme cohort-liability decomposition as well.

Economically, a negative cohort sponsor-solvency margin means the cohort-attributed residual basis has gone below zero in at least one outcome branch even though the pooled sponsor side still has enough non-fee assets to honor the aggregate winner reserve. In other words, the pool as a whole is still senior-safe, but the particular decomposition of pooled sponsor risk into cohort-level balance sheets has started to assign more outcome-contingent loss to one cohort than its own attributed basis can absorb. Figure 7 and Table 7 make that boundary concrete on one representative 64-outcome path. The exact-model reserve margin stays positive from bootstrap through the winner claim step, while the minimum cohort margin compresses from +60.54 before resolution to −189.46 at Claim Win. Layer C follows the same shape closely, which is why the paper treats this as a boundary in cohort attribution rather than as immediate user-facing winner insolvency. It does not refute the ordinary-regime

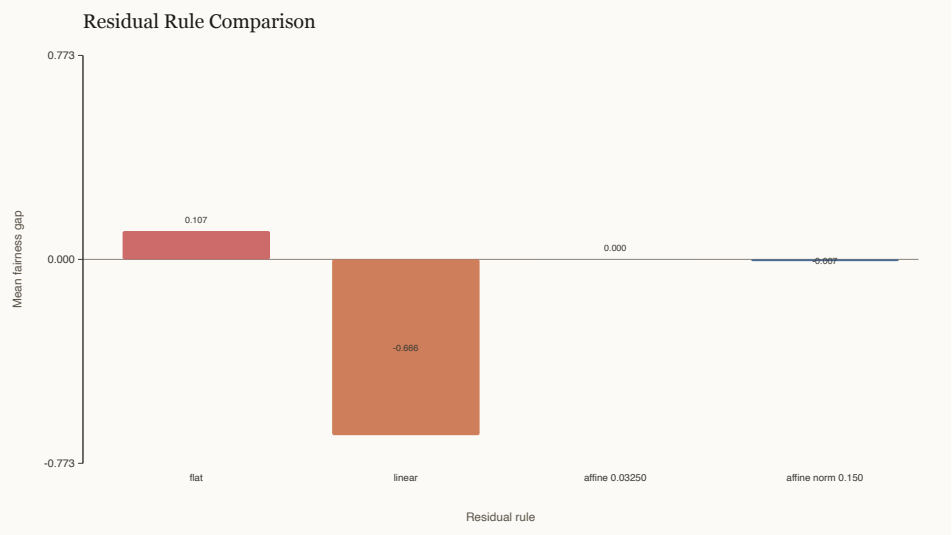

Figure 6. Residual-rule comparison across the main reference points studied in the paper, including the historical event-step crossover and the current normalized default.

Table 7. Representative low-tail failing path in the exact reference model.

Event	B	$p_{\min}$	$p_{\max}$	Reserve margin	Min cohort margin
Bootstrap	100	0.0156	0.0156	435.89	435.89
Buy	100	0.0133	0.1620	451.99	201.99
Buy	100	0.0045	0.6667	560.54	60.54
LP Enter	120	0.0045	0.6667	668.64	60.54
Resolve	120	0.0000	1.0000	168.64	60.54
Claim Win	120	0.0000	1.0000	168.64	-189.46
Claim LP	20	0.0000	1.0000	108.11	0.00
Claim LP	0	0.0000	1.0000	0.00	0.00

result, but it does tell the protocol to ship with explicit guardrails instead of assuming universality.

6 What the Results Mean

The central interpretive lesson is that fee fairness and residual fairness should not be treated as one problem. Fee fairness is the strong, clean guarantee. It can be stated crisply and implemented with a cumulative fee index. Residual fairness is the subtler problem because residual value is junior, path-dependent, and sensitive to when capital arrived relative to how uncertainty evolved.

That distinction clarifies what the active-LP model actually offers. It is not a passive yield primitive in the ordinary DeFi sense. An LP is underwriting a market maker with contingent liabilities and path-dependent residual economics. Later LPs are not receiving mysterious windfalls from nowhere. They benefit from a specific timing advantage. Once that advantage is modeled explicitly, it can be corrected rather than hand-waved.

That is also why the standard DeFi shorthand of impermanent loss is not the right headline risk description here. The active-LP mechanism does not mainly expose participants to withdrawable pool-inventory drift against a simple hold benchmark. It exposes them to a locked junior residual claim whose value depends on realized fee flow, the path of uncertainty resolution, and where the LP entered relative to that path.

The difficult part appears calibratable, but it is not reducible to a slogan such as “LPs are just buying

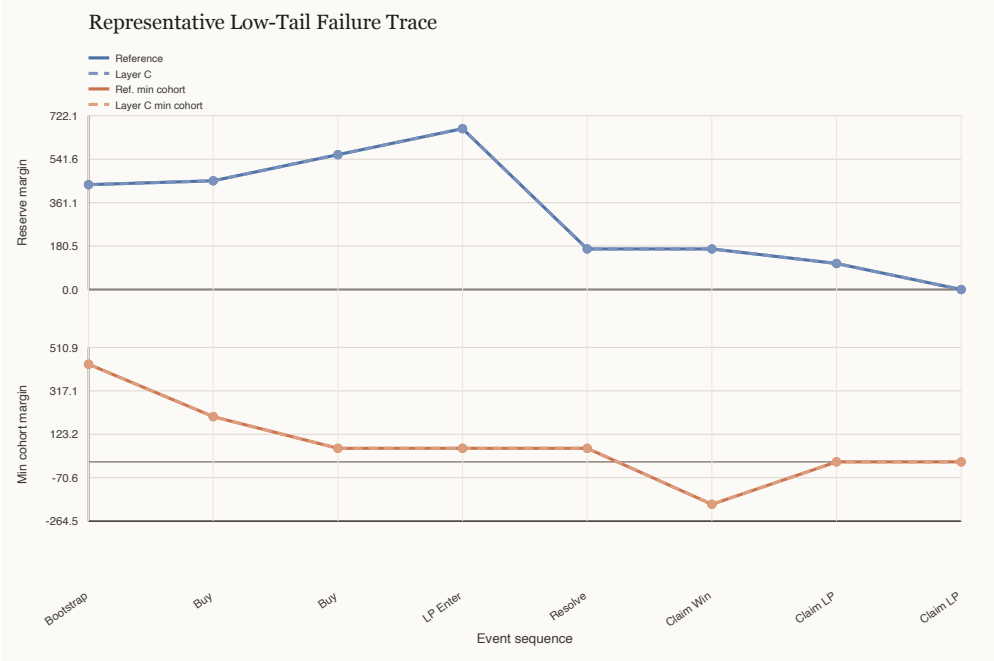

Figure 7. Representative low-tail failure trace. Aggregate reserve coverage remains positive in both the exact and Layer C models, while the cohort sponsor-solvency margin turns negative at Claim Win.

the market on both sides." The mechanism needs explicit settlement seniority, reserve-based release, and calibrated residual weighting. That is the paper's real claim: live liquidity only works once the accounting is honest.

The same point applies to participation. No-withdrawal LPs are not being offered a frictionless passive strategy. They are taking a lockup in exchange for a claim on future fees and a junior share of residual value. The mechanism works if that profile is attractive to market creators, specialized LPs, or informed capital in markets where volume and fee flow justify the lockup. It is not intended to make every market equally attractive to every LP.

7 Protocol Implications

First, state semantics must be split cleanly. Pricing state cannot double as trader-claim state. Second, active LP withdrawal should remain out of scope for the studied mechanism. Third, reserve-based settlement should be preferred over an all-winners-must-claim unlock rule because the latter makes LP capital operationally impractical. Fourth, the residual-weight parameter should remain configurable in the protocol until production hardening is complete, even if the paper now uses $\lambda_{\text{norm}} = 0.150$ as the best current normalized default. In a production protocol, that parameter should stay tied to an explicit normalized market clock rather than be reinterpreted as a raw event-step constant. Fifth, the protocol should document an explicit operating envelope and a clear parameter-governance policy, because LPs are locking capital against a rule they need to expect will remain stable. Sixth, sponsor-share issuance should stay linear in Δb if fee rights and residual weights are meant to track the same depth unit that the entry is actually adding.

The implementation path therefore has a defined target model, with a small number of configurable parameters rather than one frozen canonical setting.

Table 8. Layer C comparison across operating regimes.

Metric	Ordinary target regime	Low-tail regime
Result count	194	80
Price continuity pass rate	1	1
Slippage pass rate	0.9948	1
Reserve-coverage pass rate	1	1
Validation / diagnostic failure count	0	8
Mean scenario max quote divergence	0.0001669	0.0199139
Max scenario max quote divergence	0.0009064	0.2726206
Mean scenario max NAV/deposit divergence	0.0004717	0.0005046
Max scenario max NAV/deposit divergence	0.0181318	0.0304545

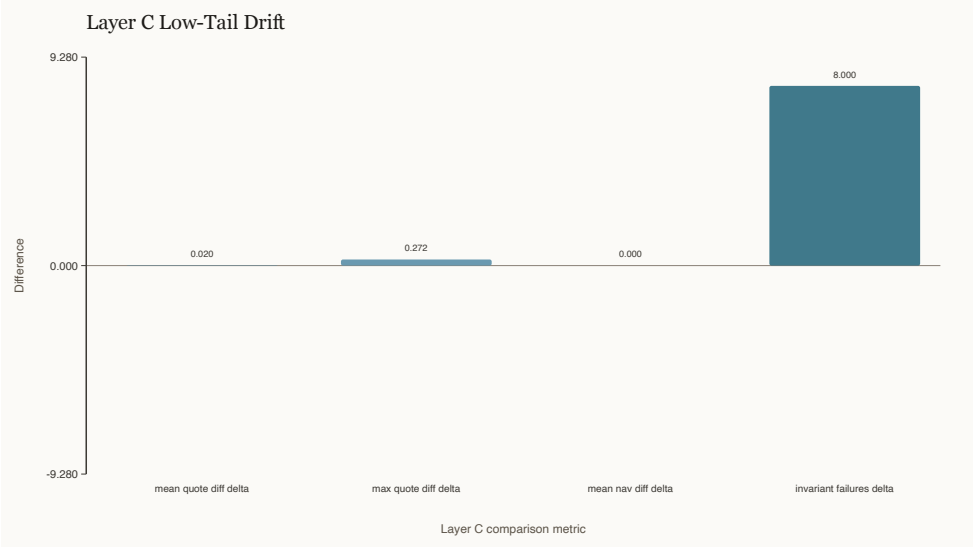

Figure 8. Layer C comparison. Ordinary regimes remain close to the reference model; low-tail many-outcome regimes define the main boundary.

8 Deployment Boundaries

The residual-weight parameter is empirical, not closed-form. The current claim is therefore bounded: within the regimes validated by the simulation program, especially the ordinary Layer C packs, the mechanism is strong enough to support implementation. The low-tail many-outcome regime remains the main boundary, so the protocol should ship with explicit limits there rather than implying universal safety.

The simulator is broad but not exhaustive, and implementation still has real state-growth, box-management, and opcode-budget work ahead. Position transfer is also constrained more tightly than in ordinary tokenized trading: either trader claims must remain non-transferable during the market, or any transfer must carry its remaining cost basis so refunds and cancellation accounting stay well-defined. Those are deployment constraints around a mechanism with a clear economic backbone, not signs of economic incoherence.

One practical implication is that active LP entry should likely be hard-gated, or at least tightly constrained, in highly skewed markets rather than treated as universally available. If the winning outcome is already near-certain, a late LP may be posting collateral into a market with limited residual upside and substantial downside to being junior. The calibrated high-skew boundary pack supports that

reading directly: once maximum entry-time probability exceeds 0.8, the mean late-minus-early gap is about −0.2222, and above 0.9 it falls to about −0.3015. The current paper does not fix a single production threshold, but it does support the idea that deployments should enforce explicit entry limits when the market approaches near-resolution states.

These boundaries define the regime in which the product should claim confidence.

9 Conclusion

The results support a clear conclusion. Active LP entry in LMSR prediction markets appears viable, but only under a narrower sponsor-style mechanism than ordinary DeFi LP intuition suggests: live entry, no active withdrawal, prospective-fee fairness, fixed-claim winner seniority, reserve-based residual release, and mild affine residual weighting on normalized lifetime exposure.

The central economic issue is residual timing bias. In this draft, that bias is measurable and materially reducible rather than fatal. On held-out ordinary-regime packs, $\lambda_{\text{norm}} = 0.150$ keeps the mean late-minus-early gap near zero and the duration-bucket drift small, while the best event-step family remains badly clock-sensitive. High-skew near-resolved states reverse the sign and justify entry gating, and the low-tail many-outcome slice remains the main cohort-attribution boundary.

What remains is concrete rather than open-ended: encode the validated design into contracts, define the operating envelope explicitly, and harden the low-tail boundary at the implementation layer.

A Reproducibility

All simulation code, traced outputs, paper artifacts, and reproducibility instructions for this manuscript are intended to live in the public repository:

https://github.com/qmrkt/active-lp

That repository is the canonical source for the empirical claims in this paper. It will contain the simulation engines, scenario definitions, artifact-generation scripts, calibration runs, and the figure/table assets used by the manuscript. The intended reproducibility standard is straightforward: each major empirical claim in the paper should be traceable to a preserved run family or derived artifact in that repository, and the paper-facing figures and tables should be regenerable from the published code and outputs without relying on unpublished local files.

For the purposes of this first draft, the appendix therefore points to the public repository rather than enumerating local workspace paths. Final publication versions should cite a stable repository revision, release tag, or archival snapshot so the evidence base remains fixed.

Acknowledgments

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